

Problem Set 1: Thinking like an Economist

Questions and Solutions

- 1) You're at your local lunch restaurant. All lunch specials cost \$15 and you have a gift card for \$15 for this restaurant that expires today. You eat only 1 dish for lunch. The menu, with your benefit for each option, expressed in dollars, is given in the table below.

Lunch Special	Your benefit
Vegetarian	\$18
Beef	\$25
Chicken	\$20
Fish	\$14

- 1a) What is the (opportunity) cost of choosing (buying and eating) the beef special?

Solution: We follow the same process as we always do.

- Step 1: Identify the resources being used for the action/choice.
- Step 2: Identify the alternatives.
- Step 3: Identify the net benefit from the next best use.
- Step 4: Calculate the OC of A as \$Value of Resources for A + Net benefit from option B
 - If Option A uses extra resources (or different resources) than Option B, make sure to value those resources at their next best use.

The resource used to buy the beef special is the gift card.

Alternatives?

- You can only use it in the restaurant so the other choices in the table are your only options.
- You only buy 1 dish, so every option except the beef is an alternative.
- The gift card expires today, so saving it for another day is not an option.

OK, so the alternatives are Vegetarian, Chicken and Fish. Each of them gives us some benefit (say B). But what resource do they use? The same gift card, with the same restrictions. That makes the net benefit from each option B minus the gift card. All of them use the same gift card and cost exactly the same, so to pick the best alternative we are looking for the one with the highest B, which is Chicken at \$20.

The OC of beef is therefore Gift card + \$20 - Gift card = \$20. **The OC of beef is giving up the option to eat chicken, which you would value at \$20.**

- 1b) What is the opportunity cost of choosing the vegetarian option?

Solution: Follow the same steps as above until you get to the alternatives. To pick the best alternative, we are looking for the one with the highest benefit, which is Beef at \$25.

The OC of the vegetarian option is therefore Gift card + \$25 - Gift card = \$25. **The OC of the vegetarian option is giving up the option to eat beef, which you would value at \$25.**

- 1c) What do you choose to eat?

Solution: The (opp.) cost of each option is given in the table below, using the original benefits from the question

Lunch Special	Your benefit	OC
Vegetarian	\$18	\$25
Beef	\$25	\$20
Chicken	\$20	\$25
Fish	\$14	\$25

Beef is the only option for which the benefit is higher than the (opp.) cost, so you choose Beef.

- 1d) You read an article on health that says fish and vegetables are healthier options. It changes your benefit from the vegetarian option to \$19 and from the fish to \$23. What is the cost of choosing the beef option now?

Solution: Chicken is no longer the next best option; that is now fish at \$23. Following the same process as above, **the OC of beef is \$23.**

- 2) You are choosing how many hours to spend studying this weekend. Your total benefit for a given number of hours spent studying is given in the table below. If you don't study, you work as a babysitter for \$20 an hour.

No. of hours	Total Benefit
1	\$67
2	\$126
3	\$177
4	\$220
5	\$255
6	\$282
7	\$301
8	\$312
9	\$315

- 2a) How many hours do you study this weekend?

Solution: 6 hours – the sixth is the last hour whose marginal benefit still covers the \$20 you give up in babysitting wages.

Whenever you see a “how many?” question, your first instinct should be to calculate the marginal cost and marginal benefit of an extra unit. Here your units are hours.

For marginal benefit, we want the extra benefit from each additional unit. For any quantity Q , the marginal benefit of the Q^{th} hour is the total benefit when you have Q hours, minus the total benefit from when you have $Q - 1$ hours. The table below does this calculation for you.

No. of hours	Total Benefit	Marginal Benefit	Marginal cost	Total Cost
1	\$67	\$67	\$20	\$20
2	\$126	\$59	\$20	\$40
3	\$177	\$51	\$20	\$60
4	\$220	\$43	\$20	\$80
5	\$255	\$35	\$20	\$100
6	\$282	\$27	\$20	\$120
7	\$301	\$19	\$20	\$140
8	\$312	\$11	\$20	\$160
9	\$315	\$3	\$20	\$180

The marginal cost of an hour spent studying is the money you could have earned if you spent that hour in your next best alternative, i.e. babysitting, which is \$20.

For the first 6 hours, the marginal benefit is higher than the marginal cost. For the seventh hour, your additional benefit is \$19, but you are giving up \$20 in babysitting wages, so you study only 6 hours – IF you study at all.

Should you study at all? For that, let's compare the Total Benefit (TB) and Total Cost (TC). $TB = 282$, $TC = 20 * 6 = 120$. As $Total\ Benefit > Total\ Cost$, you choose to study.

- 2b) Suppose now your Total Benefits are given by the revised table below. With the babysitting outside option of \$20, how many hours will you study?

No. of hours	1	2	3	4	5	6	7	8	9
Total Benefit	\$39	\$68	\$93	\$104	\$139	\$168	\$193	\$204	\$207

Solution: In the same way as in the previous question, create the table below. Up to the third unit $MB \geq MC$. Going to the 4th reverses this, so we'd like to stop here. But notice these are not well-behaved MB numbers and in cases like these we want to look around and ensure we're not missing anything. If we move to the 5th unit, the MB is higher than the MC and more than enough to compensate for the next loss from the 4th unit. This is why we want to continue till the largest unit where $MB \geq MC$ with the caveat that we need to make sure that at this quantity $TB \geq TC$. If not, we stop at the previously identified quantity where $MB \geq MC$. The largest number of hours for which $MB \geq MC$ is 7 hours. At 7 hours we have $TB > TC$, so we choose 7 hours. If this was not the case, we would go back and check if they wanted to stop by a quantity of 3. Here it is better to continue till 7 and you can verify this. If we stopped at 3 hours, we would get $TB = 93$, $TC = 60$, a net benefit of \$33. **If we choose 7 hours we get a net benefit of \$53, so 7 hours is the better choice.**

No. of hours	Total Benefit	Marginal Benefit	Marginal cost	Total Cost
1	\$39	\$39	\$20	\$20
2	\$68	\$29	\$20	\$40
3	\$93	\$25	\$20	\$60
4	\$104	\$11	\$20	\$80
5	\$139	\$35	\$20	\$100
6	\$168	\$29	\$20	\$120
7	\$193	\$25	\$20	\$140
8	\$204	\$11	\$20	\$160
9	\$207	\$3	\$20	\$180

- 2c) With the revised Total Benefits table in (2b) your babysitting wage increases to \$30 an hour. How many hours will you study?

Solution: The revised table with the new costs is below. The largest quantity for which $MB \geq MC$ is 5 hours, but at 5 hours we have $TC > TB$, so it is not an option: net benefit is negative. If we stop earlier, at 1 hour, we have $MB \geq MC$ and $TB \geq TC$ both satisfied, so **if the wage rises to \$30 you would study for just 1 hour.**

No. of hours	Total Benefit	Marginal Benefit	Marginal cost	Total Cost
1	\$39	\$39	\$30	\$30
2	\$68	\$29	\$30	\$60
3	\$93	\$25	\$30	\$90
4	\$104	\$11	\$30	\$120
5	\$139	\$35	\$30	\$150
6	\$168	\$29	\$30	\$180
7	\$193	\$25	\$30	\$210
8	\$204	\$11	\$30	\$240
9	\$207	\$3	\$30	\$270

2d) Suppose instead of the table, you were given an equation that summarized your marginal benefit of studying for the Q th hour as $MB(Q) = 75 - 8 * Q$. What is your benefit from the 3.5th hour spent studying over the weekend? Use the equation.

Solution: \$47, from plugging 3.5 into the $MB(Q)$ equation to find that $MB(3.5) = 75 - 8 * 3.5 = 47$. The marginal benefit at the instant you study for 3.5 hours is \$47.

2e) If your Marginal Benefit from studying is summarized by $MB(Q) = 75 - 8 * Q$, how many hours do you spend studying over the weekend?

Solution: 6.88 hours, the point where MB = MC. 6.875 rounded to 2 decimal places becomes 6.88. In 101, fractions of units are always possible with equations because we are given information for tiny bits, implicitly assuming that fractions are possible. Always round to two decimal places.

Because we assume that hours of studying is a continuous variable, you study until marginal benefit is exactly equal to marginal cost. With discrete data this is not possible as we don't have information for the jumps between integers and have to make do with the largest quantity where $MB \geq MC$.

$$\begin{aligned}
 \text{marginal benefit} &= \text{marginal cost} \\
 75 - 8Q^* &= 20 \\
 55 &= 8Q^* \\
 Q^* &= \frac{55}{8} = 6.875 = 6.88
 \end{aligned}$$

2f) Suppose I don't know how much you get per hour babysitting, but I know that you study 7.5 hours this weekend. Assume your marginal benefits are captured by $MB(Q) = 75 - 8 * Q$. How can I calculate your hourly wage rate?

Solution: I know that at the optimal quantity Q^* , your $MB(Q^*) = MC$. If you study 7.5 hours, then it must be that at 7.5 hours

$$\begin{aligned}
 \text{marginal benefit} &= \text{marginal cost} \\
 75 - 8(7.5) &= MC \\
 15 &= MC
 \end{aligned}$$

Your hourly wage must be \$15, since that is the marginal cost that makes 7.5 hours the optimal choice. I can check this by working out what you would choose if your MC were \$15.

$$\begin{aligned}
 \text{marginal benefit} &= \text{marginal cost} \\
 75 - 8Q^* &= 15 \\
 60 &= 8Q^* \\
 Q^* &= \frac{60}{8} = 7.5
 \end{aligned}$$

- 2g) Suppose your Marginal Costs are now $MC(Q) = 9 + 4Q$ and Marginal Benefits are $MB(Q) = 75 - 8Q$. Find your optimal hours of studying.

Solution: To find the optimal hours of studying, set $MB = MC$ as below.

$$\begin{array}{rcl} \text{Marginal Benefit} & = & \text{Marginal Cost} \\ 75 - 8Q^* & = & 9 + 4Q^* \\ 66 & = & 12Q^* \\ Q^* & = & \frac{66}{12} = 5.5 \end{array}$$

You choose to study for 5.5 hours, the point where MB and MC are equal.

- 3) You go through a lot of socks a month. You notice an advertisement for a sock club from your favorite sock company. You pay a monthly membership fee of \$M and then you can buy any number of socks you want for \$5 each. Your marginal benefit in dollars, as a function of Q (i.e. pairs of socks in a month) is $MB(Q) = 20 - 5Q$.
- 3a) You are offered a discounted rate and the monthly fee is \$15 (i.e. $M=15$). Do you purchase a membership? If so, how many pairs of socks do you optimally purchase? (Hint: First, assume you are a member, and then figure out the quantity you would buy. You would only buy the membership if the total benefit is at least as large as the total cost.)

Solution: Step one is to solve the “how much?” part. IF you bought the membership, how many pairs would you buy?

To find this, we set $MC=MB$.

Each pair costs \$5 so that is your marginal cost.

$$\begin{array}{rcl} \text{Marginal Benefit} & = & \text{Marginal Cost} \\ 20 - 5Q^* & = & 5 \\ 15 & = & 5Q^* \\ Q^* & = & \frac{15}{5} = 3 \end{array}$$

Now, decide if you will buy the membership or not, given that IF you bought it, you would buy 3 pairs of socks.

After you pay the membership fee, the benefit from the membership is your marginal benefit added up over every tiny bit of extra sock.

Graphically, it is the shaded area in the figure below plus the rectangle below it: the area under the MB curve for the 3 units you buy, IF you bought the membership.

Why this area? Remember, equations give us your MB for tiny bits of extra socks. We want to calculate the MB for every tiny bit of extra socks and then sum it up to get total benefit. When you have an equation, you have the information on MBs to do that and so you use areas. When you are artificially given integer values, then you can only add up the MB for the integer values. Is thinking you can consume 0.01 socks weird? Not if you think of it as thousands of socks, or average socks per month or week.

We can calculate the area as $\frac{1}{2} * 3 * 15 + 3 * 5 = 22.5 + 15$.

The total benefit of the membership is \$22.5+\$15 and the total cost is \$15+\$M. **With M = \$15, benefits are larger than costs, so you buy the membership.**

- 3b) Next month, the discount no longer applies and the monthly rate is \$25. Do you buy the membership?

Solution: No, because the total benefit of the membership is \$22.5+\$15 and the total cost is \$15+\$M. **With M = \$25, benefits are smaller than costs, so you do not buy the membership.**

Notice that since the MC is still \$5, your quantity choice would not change IF you bought the membership.

- 3c) Unfortunately, you did not read the fine print when you signed up and you have to pay for an additional month before you can cancel the membership. How many pairs of socks do you buy in the second month?

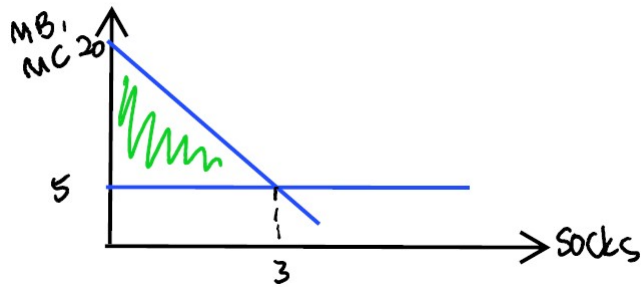


Figure 1: Marginal benefits and costs of club

Solution: There is nothing you can do about recovering your membership fee. You are a member no matter what. You don't have the option of not being a member. Now that you are a member, you compare only MB and MC to choose your quantity. **As the marginal benefits and marginal costs haven't changed, you still buy 3 pairs of socks.**

3d) After a quick call to customer service, they allow you to cancel your membership in the second month but they only refund you \$20 of the \$25 you paid as membership fee (they say it is for processing charges). Do you accept their offer and cancel your membership for the second month?

Solution: If you cancel your membership, only \$20 of your membership fee is recoverable as the rest is sunk. If you keep the membership:

Total Benefits: \$37.5

Total Cost: $\$20 + \$15 = \$35$. The socks cost \$15 and only the non-sunk portion of the membership fee (i.e. \$20) is relevant.

As total benefits are higher than total costs, you don't accept the refund: you keep the membership for the second month and cancel from the third month onward.

3e) Suppose you did not cancel and have a membership for the second month. Your friend looks at your shopping cart and encourages you to buy more than 3 pairs. Should you listen to them?

Solution: No, you should not listen to them: if you buy the extra pair, your marginal benefit is less than your marginal cost for that pair. This reduces your net benefits from the membership. Similarly, if you bought fewer than 3 pairs, you would be missing a pair for which the marginal benefit is higher than the marginal cost. That would also reduce the net benefits from your membership.

4) I have a credit at the iTunes store expiring today. I have narrowed down my choices to the albums in the table below, which also contains the dollar value of my enjoyment for each. Suppose that each album costs \$10. I don't want to spend any extra money on music this month.

Artist	Album	Value
Jidenna	85 to Africa	\$11
Lauryn Hill	Miseducation of Lauryn Hill	\$13
Amy Winehouse	Back to Black	\$15
Zaz	Effet miroir	\$17
Lizzo	Cuz I love you	\$19

4a) I have \$20 in credit. What albums would you predict I put in my shopping cart?

Solution: I will choose the ones with the highest value to me, so the Lizzo and Zaz albums go in my shopping cart.

- 4b) I find a birthday gift card I forgot about (also expiring today) for an additional \$10. What will I add to my shopping cart?

Solution: I will add the next album in my ranking, Amy Winehouse's album.

- 4c) Right before I click buy, I notice that Taylor Swift's new album is available for purchase today. If I add it to my shopping cart, what do you know about my minimum benefit from her album?

Solution: I would give up the lowest ranked (by me) album in my cart (Winehouse) if I chose Taylor Swift's album, so **the minimum I must value her album at is \$15**. From the question above you know that if I had to choose only two, I would not buy Winehouse.

- 4d) Taylor Swift's album release is delayed, so that is no longer an option. If instead of a gift card I had \$10 in cash that I could spend on music. What albums would you predict I buy?

Solution: In this case, I would need to know what else I could spend that \$10 on. If I had decided to spend it only on music and nothing else, the logic would be exactly the same as before. The value from the highest ranked album is \$19. The only other alternatives are the other albums and the next best alternative is Zaz which gives us an OC of

$$\$10 + \$17 - \$10 = \$17$$

As Benefit $\geq$ OC, I buy the Lizzo album.

But what if I could instead spend the \$10 on one bubble tea that I valued at \$20 (no half teas or fractions possible), and that is the only other option I have? Then I would calculate the OC of spending the money on music.

$$OC_{music} = \$10 + \$20 - \$10 = \$20$$

Now OC is higher than Benefits, so I would not buy any albums and would buy the bubble tea instead.

One way of thinking of a gift card is as a way to restrict options to the albums on iTunes. The other important information is that the gift card expires today. If not, and I know that Beyonce is dropping an album in the near future I may choose to keep the gift card and use it later to buy her album.

- 4e) Beyonce is coming to town and I was one of the lucky few to get a ticket to see her for \$200. The day of the show, tickets are selling online for \$320. What is my Opportunity Cost of going to the concert? Note: I'm leaving this open-ended, because I want you to think about what information you need, rather than just memorize patterns of where to plug in what given number.

Solution: If I go to the concert, I use up cash resources of \$320. **Why not \$200? Because the \$200 is sunk.** I can't get a refund from the ticket company (nor do I want to, even if I could), but I could sell the ticket online. If I see Beyonce I am giving up the chance to earn \$320 by selling my ticket. I am also using up my time to attend her concert. Say it takes 3 hours. In that case, I'm also losing the value of 3 hours spent doing something else, say laundry, which doesn't take any additional resources. If laundry took any additional resources, then I'd have to factor them in as well. The OC of going to see Beyonce is

$$OC_{Beyonce} = \$320 + \$V_{3hrs}$$

- 5) You are going to a movie, which costs \$15 and takes up 2 hours of your time.

- 5a) What is the opportunity cost of going to the movie?

Solution: We follow the same process as we always do.

- Step 1: Identify the resources being used for the action/choice.
- Step 2: Identify the alternatives (below assumes you've already done that)
- Step 3: Identify the net benefit from the next best use.
- Step 4: Calculate the OC of A as \$Value of Resources for A + Net benefit from option B
 - If Option A uses extra resources (or different resources) than Option B, make sure to value those resources at their next best use.

The resources used in going to a movie are

- 2 hours of time
- \$15 cost of ticket

What is the next best alternative? Suppose it is studying, which also takes exactly 2 hours of your time but is free otherwise. The net benefit is the value expressed in dollars of the time you spend studying. Denote it as $\$V_{studying}$. Since there are no other resources, that is your net benefit.

This makes the opportunity cost of the movie $OC_{movie} = \$15 + \$V_{2hrs} + \$V_{studying} - \$V_{2hrs} = \$15 + \$V_{studying}$.
Note: if the question does not specify how you value spending the \$15, you can assume that its value is \$15. E.g. here, since it is not specified otherwise, you could write it as

$$OC_{movie} = \$15 + \$V_{studying}$$

5b) What is your willingness to pay (WTP) for the movie? Write an equation in terms of WTP that determines whether you go to the movie or not.

Solution: Economists use the general rule that people should buy the good as long as Benefits $\geq$ Opportunity Costs. But when you buy something, you typically compare WTP to price. We can move things around to convert the Benefits $\geq$ Opportunity Costs into a WTP $\geq$ Price comparison.

For example, in this case you go for the movie if

$$\$V_{movie} \geq \$V_{studying} + \$15$$

Keep the \$15 on the right-hand side and move everything else to the left-hand side of the equation

$$\$V_{movie} - \$V_{studying} \geq \$15$$

Rename what is on the left-hand side as WTP. In this case it would be

$$WTP = \$V_{movie} - \$V_{studying}$$

You go for the movie if WTP $\geq$ Price. Which in this case would be

$$WTP \geq \$15$$

OK, so that is still just algebra. What does WTP mean? Let's look at it again.

$$WTP = \$V_{movie} - \$V_{studying}$$

$\$V_{movie} - \$V_{studying}$ is the extra value you get from watching a movie over and above your next best option.

If that is above the price, you go to the movie; otherwise you spend the time studying.

6) A new textbook costs you \$100 while a used one costs you \$40.

6a) What is the opportunity cost of a new textbook?

Solution: The value of the next best option for each resource (expressed in \$s) is the opportunity cost of that action/choice.

The resources used in buying a textbook are cash (i.e. \$100).

If you don't buy a new textbook, the next best alternative is to buy a used textbook. The resources used in that case are \$40. What is the benefit of the used text? We haven't been given that number, so let's use a stand-in ($\$V_{used}$) to represent the value to you of a used textbook converted into \$s.

The net benefit of a used textbook is $\$V_{used} - \40 .

This makes the opportunity cost of a new text book

$$OC_{new} = \$100 + \$V_{used} - \$40 = \$V_{used} + \$60$$

The opportunity cost of a new textbook is giving up the chance to own a used textbook and \$60 in your pocket to spend on something else.

Note: if the question does not specify how you value spending the \$60 you can assume that its value is \$60. e.g. here since not specified otherwise, we write it as \$60.

- 6b)** What is your willingness to pay (WTP) for the new textbook? Write an equation in terms of WTP that determines whether you buy the new textbook or not.

Solution: Economists use the general rule that people should buy the good as long as Benefits $\geq$ Opportunity Costs. But when you buy a textbook, typically you compare WTP to price. We can move things around to convert the Benefits $\geq$ Opportunity Costs into a WTP $\geq$ Price comparison.

For example, in this case you would buy the textbook if

$$\$V_{new} \geq \$V_{used} + \$60$$

But the price is \$100. Let's do some algebra to get the price in there. Add and subtract \$100 on the right-hand side of the equation

$$\$V_{new} \geq \$V_{used} + \$60 - \$100 + \$100$$

Keep one \$100 on the right-hand side and move everything else to the left-hand side of the equation

$$\$V_{new} - \$V_{used} - \$60 + \$100 \geq \$100$$

Rename what is on the left-hand side as WTP. Here it would be

$$WTP = \$V_{new} - \$V_{used} + \$40$$

You buy the new book if WTP $\geq$ Price. Which in this case would be

$$WTP \geq \$100$$

OK, so that is still just algebra. What does WTP mean? Let's look at the parts separately.

$$WTP = \$V_{new} - \$V_{used} + \$40$$

- $\$V_{new}$: The benefit you get from the new textbook is one component. That should make sense. If something gives you a benefit, you are willing to pay up to that benefit for it. But...
- You could get $\$V_{used} - \40 amount of net benefit from the alternative.
- **The word net matters: you are willing to pay up to the extra value you get over and above the net benefit from your next best alternative. That is your WTP.**

- 7)** You work for a district government in a developing country as an economist. The district has good schools, and education is free. However, the average number of years of schooling children complete is low. Through this question, let's see if we can use opportunity cost to understand why this could be the case.

- 7a)** This district is an agricultural district and suppose for simplicity, that everyone works on a family farm. If kids don't go to school, they work on the family farm. For a family that has a kid that could attend school,

1. what is the cost of one year of schooling for that kid? Remember, education is free so you would have to think of opportunity cost. You could consider things like the cost of books, uniforms, etc., but let's assume those away for now.

Hint: What is the resource the kid is using to attend school and what is the value of that resource in the next best alternative use, working on the farm?

2. Is this a marginal cost or a total cost?

Hint: Is this a cost for an extra something or just comparing doing X or not?

3. Does it depend on the age of the kid?

Hint: How does the value of the kid's time working on the farm depend on their age?

Solution: If education is free, then the resource used by the family for the kid to go to school is just the kid's time. If the kid spends 9am-3pm in school, they're taking that time away from the farm. **The value of this time is the family income the kid would have contributed by working on the farm instead.**

If we think of this as an extra year of school and frame the decision of "how many" years of education, then it would be a marginal cost: the cost of an extra year of schooling. If we think of it instead as a yes-or-no choice about 5 years of school, we would still have to work out where the number 5 came from, and that would involve thinking about how much an extra year costs.

Note: If you want to think about it as an extra day of school etc. you would work through it in a similar fashion.

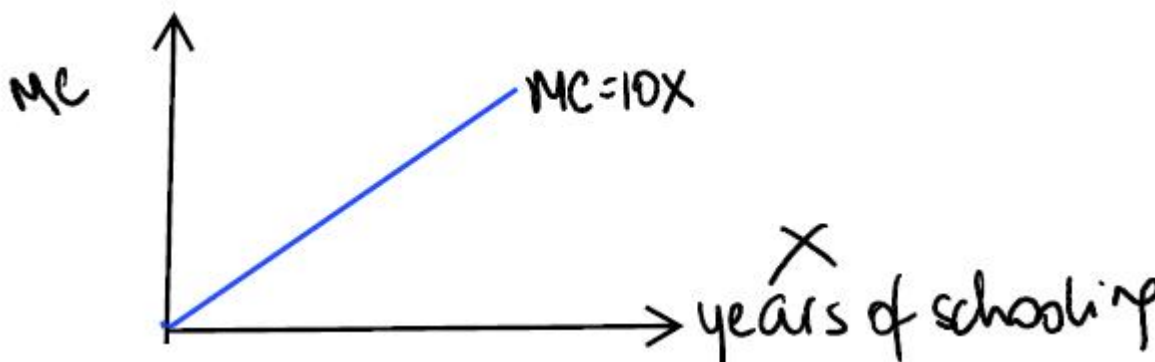
The age of the kid may affect the opportunity cost. For example, the older the kid, the stronger they are and the more they contribute to the household.

7b) Suppose the extra cost to the family for the X th year of schooling is $\$10X$. What is the marginal cost function for schooling? Explain and draw it on a graph.

Solution: The key here is the word extra in the description. The **extra** cost to the family when the kid goes from $X - 1$ to X years of schooling is $\$10X$. Although it might look like the Total Cost functions elsewhere in the problem sets, the wording tells us this is a marginal cost.

This makes

$$MC(X) = 10X$$



7c) Education is beneficial. With an extra year of education, the kid is more productive and gets paid a higher wage. The *total* income the kid earns in the future for X years of education is $\$80X$.

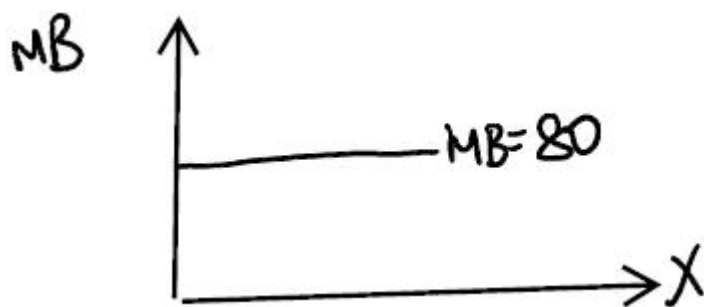
What is the marginal benefit of an extra year of schooling? Explain and draw it on a graph.

Note: kids only earn this wage as adults, whereas the income the family loses due to school attendance is incurred now.

This is tricky because parents pay the cost now but kids earn the income in the future. Suppose that parents are altruistic and care about the kid's future income in the same way the kid would. That makes the argument simpler, as the kid's future wage is the parents' benefit now, when they are making the choice.

Solution: Again, read the question carefully. The **total** income earned with X years of education is $\$80X$. This means that going from $X-1$ to X years of schooling gives the kid

$$MB(X) = 80X - 80(X - 1) = 80$$



Note: if you find this difficult to understand, you may need to revisit your high school math. Try the ungraded math assessment quiz on Quercus

- 7d) Given the information in parts (b) and (c), how many years of schooling will a kid in your district get? Explain and draw it on a graph.

Solution: This is a 'how much' question. To get an extra year of schooling, the kid (or altruistic parent) will compare the MC and MB of that extra year. If MB is bigger than MC, stay in school. If not, leave. At the point where they are equal, the kid is indifferent, and that is the stopping point.

$$MB = MC$$

$$80 = 10X$$

$$\Rightarrow X = 8$$

The kid will get 8 years of education, the point where the marginal benefit of \$80 is just matched by the marginal cost.

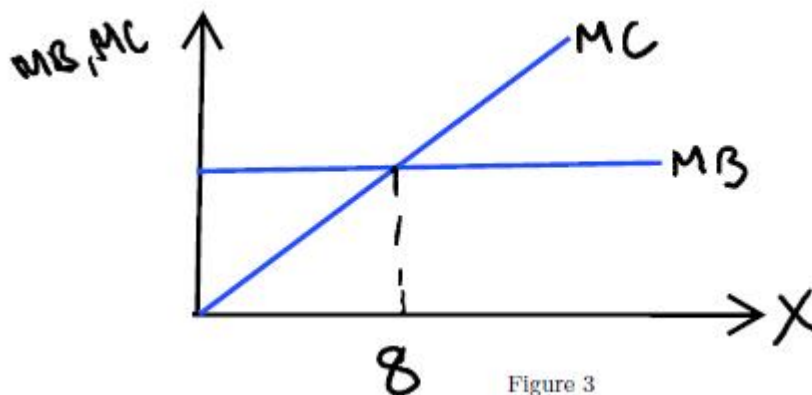


Figure 3

- 7e) A BPO (business process outsourcing) center opens in your district. Jobs there pay well. The *total* income the kid earns in the future for X years of education is \$100X. Does this change the number of years of schooling a kid in your district gets? Explain and draw it on a graph.

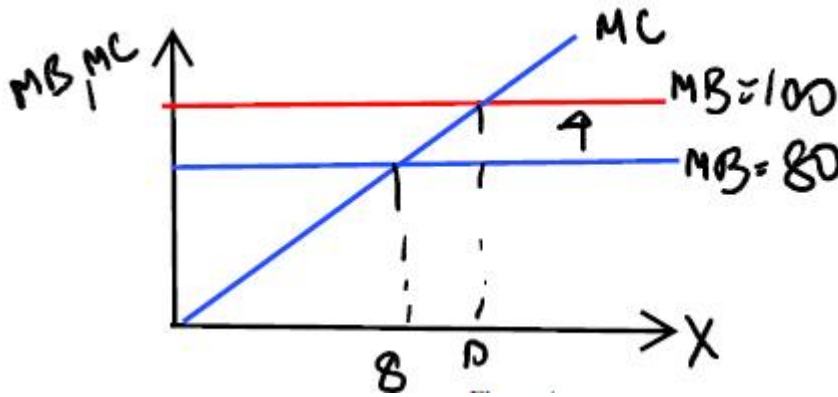
Solution: This raises the MB of education to \$100. Stopping at 8 years is no longer optimal, because at 8 years the marginal benefit is \$100 but the marginal cost is only \$80. The kid will add another year, and will keep adding years of education until

$$MB = MC$$

$$100 = 10X$$

$$\Rightarrow X = 10$$

The kid will get 10 years of education with the new job opportunities.



7f) Can you use your answers above to explain why students in your district have a low average years of schooling?

Solution: A combination of high OC (kids' labour is valuable on the farm) and low outside opportunities (they work on the farm instead of at a BPO).

8) Your friends got engaged last weekend. They ask you to help them decide how much to spend on their wedding. Guide them through the process by making a table similar to the ones in the questions above. Think carefully about how you would explain the concepts of opportunity cost, marginal benefit/cost vs total benefit/cost in simple English. Why will they need to monetize the value of each option to calculate the implicit cost, and how can they incorporate intangible feelings into this process?

Question inspiration: “[Financial planning: The opportunity cost of a lavish wedding](#)” by Adhil Shetty in the Financial Express August 5, 2019.

Solution: First, let's clarify what your friends are asking you for advice on. This is a “how much” question. That means we need to get them to think of this as should they spend the *extra* dollar on wedding expenses or not.

The Marginal Cost of an extra dollar is not just \$1, but what they are giving up by putting that dollar towards their wedding, the Marginal Opportunity Cost (OC). Here they will have to think carefully about what they are reallocating the money away from. Is it retirement savings, a vacation, loan payments?

Will the Marginal (opp.) Cost increase as they spend more money on their wedding? Can you explain why? Maybe they reallocate money away from less important uses first, but as they spend more, the money has to come from more important priorities, where spending less would hurt more.

What about Marginal Benefit? One extra dollar means they can have a bigger and better wedding. How would they quantify this benefit in dollars? They spend the first few dollars on necessities for a ceremony, so their Marginal Benefits (MB) are higher. Does the MB decrease as they spend more money?

Questions for you to think about: Why would one person have a higher benefit from using the wedding money for retirement vs. spending it on a holiday? What role do preferences play here?

What role does income play here? If your friends were investment bankers instead of graduate students on a teaching assistant salary, would it make a difference to their cost/benefits? Why or why not?

9) You work for a government in a developing country as an economist. Below are some problems your colleagues are having a hard time solving and have brought you in to consult. For each, use your “thinking like an economist” skills to help them.

NOTE: This question is to give you a preview of how economists use course concepts in their research. It is OK if you can't relate Opportunity Cost and Thinking at the Margin to these questions right away. These are from academic papers, and it is difficult to make the link. I've included it to show you that even with a basic understanding of the framework you can apply it to interesting real-world questions. Think about them and then go to the solutions.

9a) [Childhood immunizations](#) are a cheap way to prevent millions of children dying each year in the developing world. Your government has made immunizations free; however, only 2% of children receive the full course of immunizations (you need multiple doses, several months apart to become immunized). A survey points to the biggest factor being the high absentee rates of health providers in village clinics with no predictable pattern. Can you explain why this would affect a parent's decision to immunize their child?

Solution: While there is no explicit cost to the vaccine, a parent still has to take time off work to take the child to the clinic. If there is no guarantee that the nurse is there, they expect to make multiple trips. A day off means lost wages, a high opportunity cost for poor low-wage workers. **A focus on explicit costs (i.e. the cost of the vaccine), while ignoring implicit (and opportunity) costs, is the problem here.** You could advise your colleagues to improve health infrastructure and ensure someone is always present to administer vaccines to improve immunization rates.

Read more about the research on this topic in the J-PAL Policy Briefcase. 2011. "[Incentives for Immunization.](#)" Cambridge, MA: Abdul Latif Jameel Poverty Action Lab.

- 9b) The district has good schools and education is free. However, there is a gender gap—girls get fewer years of education than boys. Girls leave school to get married and have kids. Your colleagues think this is because of cultural norms and there is nothing you can do to change it. Can you come up with reasons, using marginal benefits and marginal costs, why this could be the case?

Solution: This is an example of a “how much”—how many years of schooling and why does it vary by gender? The first step is to calculate the marginal benefit of an extra year of schooling. Why would an additional year of school be valuable? One reason could be higher earnings in the job market. More education → more skills → more value to employer → higher wage. On the marginal cost side, more education → higher wage if you leave school and get a job → more money forgone to stay in school → higher marginal (opportunity) cost.

One reason could be that girls have fewer job market opportunities than boys, which makes the marginal benefit of an extra year of education lower, so the stopping point comes earlier.

In the lower marginal benefit case, if you raised job opportunities for women, you should see higher education levels. This is what Robert Jensen finds: in his study, when parents were informed about job opportunities for their daughters in the BPO (business process outsourcing) industry in India, girls stayed in school, were healthier, and invested more in learning job skills.

Read more about the research on this topic in J-PAL’s “[Impact of Improved Labor Market Opportunities on Marriage and Fertility Decisions in India](#)”. It is a summary of Robert Jensen’s paper “Do Labor Market Opportunities Affect Young Women’s Work and Family Decisions? Experimental Evidence from India.” Quarterly Journal of Economics 127(2012): 753-792.

- 9c) The district government has a limited budget and has to decide between spending its money on health or education. Unfortunately, it can’t fund both. At the budget meeting, one of your colleagues makes a case for health. He says that no child should die from a preventable disease, no matter what the cost is. How would you as an economist respond to this?

Solution: Does this question make you uncomfortable? You’re [not alone](#). Our answers to these questions are one reason people dislike economists so much. This question has two parts. The first is an either-or part. Which program to fund – health or education? The second is a “how much?” part, how many dollars do we allocate to health? Both questions need a *proper* measurement, in dollars, of the marginal benefits and costs. The costs here need to include both explicit and implicit costs.

An economist in this situation would recommend calculating the opportunity cost of each option and then spending on the one with the lowest opportunity cost. Within each option, an economist would recommend spending an extra dollar as long as the marginal benefit is greater than or equal to the marginal cost. Then compare total costs to benefits once the dollar amount is decided.

Heartless, you say? Maybe. It is our, admittedly imperfect, way of making tough choices in a world of scarcity. This is why economists need and use data-driven decision making.

It is one reason behind the [randomized control trial revolution in development](#) economics. Being able to identify and quantify the causal impact of a policy helps direct resources more effectively. Read more about randomization and its use in making policy decisions [here](#) and a discussion of its shortcomings [here](#). To see what this would look like, look at J-PAL Policy Bulletin. 2017. “[Roll Call: Getting Children Into School.](#)” Figure 1 on page 9 shows you a comparison of various education interventions and their measured impact on student participation.

Checked with Claude (Anthropic) for grammar, clarity and errors. Tell me if you find one I missed.